Adaptive Resonance Theory (ART1) Practical Guide

Aim

To implement Adaptive Resonance Theory (ART1) for clustering binary input patterns using Python.

Objectives

• Understand ART1 neural network.
• Perform clustering of binary patterns.
• Learn vigilance parameter concept.
• Visualize cluster assignments.

Theory

Adaptive Resonance Theory (ART) is a neural network architecture used for pattern recognition and clustering. ART1 works on binary input patterns. The network compares incoming patterns with existing clusters. If similarity exceeds the vigilance parameter, the pattern joins the cluster; otherwise a new cluster is formed. ART solves the stability-plasticity problem by learning new patterns without forgetting old patterns.

Formula

Similarity = (Matching 1s) / (Total 1s in Input Pattern)

Vigilance Parameter

Higher vigilance creates more clusters. Lower vigilance creates fewer clusters.

Algorithm

1. Input binary patterns.
2. Initialize clusters.
3. Compare patterns with clusters.
4. Calculate similarity.
5. Assign or create cluster.
6. Display graph.

Program Code

```
# ART1 - Adaptive Resonance Theory

import numpy as np
import matplotlib.pyplot as plt

p = np.array([[1,1,0,0],
              [1,0,0,0],
              [0,0,1,1],
              [0,0,1,0]])

v = 0.6
c = []
labels = []

def sim(a,b):
    return np.sum(a & b) / np.sum(a)
for x in p:
```

```
        done = False

        for i,y in enumerate(c):

            if sim(x,y) >= v:
                c[i] = x & y
                labels.append(i+1)
                done = True
                break

        if not done:
            c.append(x.copy())
            labels.append(len(c))

    print("Cluster Assignment:", labels)

    plt.bar(range(1,len(labels)+1), labels)
    plt.xlabel("Pattern Number")
    plt.ylabel("Cluster Number")
    plt.title("ART1 Clustering")
    plt.show()
```

Code Explanation

The program imports NumPy and Matplotlib libraries. Binary patterns are defined and compared using a similarity function. Patterns with sufficient similarity are assigned to existing clusters; otherwise new clusters are formed. Finally, a bar graph visualizes the cluster assignment.

Instructions to Run in Anaconda

For .py File:
1. Open Anaconda Prompt.
2. Go to file location using cd command.
3. Run:
python filename.py

For .ipynb File:
1. Open Anaconda Navigator.
2. Launch Jupyter Notebook.
3. Open notebook file.
4. Click Run or press Shift + Enter.

Important Viva Questions & Answers

Question	Answer
What is ART1?	ART1 is a neural network used for clustering binary patterns.
What is vigilance parameter?	It controls cluster similarity matching.
What data is used in ART1?	Binary data.
What if similarity is low?	A new cluster is created.
Applications of ART?	Pattern recognition, clustering, image processing.

Outcome

Successfully implemented ART1 clustering and visualized cluster assignments.

Conclusion

ART1 effectively groups similar binary patterns into clusters using vigilance-based similarity matching.